An Adaptive Explainable Cognitive Graph Digital Twin for Cyber-Resilient Smart Hospital Energy Ecosystems

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Abstract. Smart hospital campuses rely on localized microgrids to sustain intensive-care ventilators, surgical robotics, and biomedical cold chains. A stealthy False Data Injection Attack (FDIA) on substation Remote Terminal Units can corrupt the energy-management state estimator while evading its χ^2 residual check, placing patient-critical loads at risk. We introduce a Cognitive Graph Digital Twin (CGDT) that mirrors the hospital electrical topology as a dynamic admittance graph and runs a multi-task Admittance-Conditioned GraphSAGE backbone for simultaneous attack detection and bus-level localization in a single forward pass at 0.908 ms latency. To bridge the gap between post-hoc saliency diagnostics and inductive training, our Attribution-Guided Multi-Task Loss supervises backpropagation gradients via attribution alignment to target true substation anomaly locations, driving a 5.52 percentage-point improvement in localization F1. Dynamic physical-to-virtual coherence is governed by a state-space tracking filter coupled with an adaptive exponential baseline. Upon detecting an attack, a closed-loop healing module quarantines compromised sensor channels and reconstructs measurements via graph-guided spatial interpolation, reducing Weighted Least Squares estimation error by 99.97%. On the IEEE 14-bus test feeder under full nonlinear AC power flow, the CGDT achieves $98.25\% \pm 0.34\%$ detection F1 and $97.62\% \pm 0.84\%$ localization F1 across five random seeds, and retains 92.80% detection F1 under zero-shot topology reconfigurations without retraining.

Keywords: Graph Digital Twin · Healthcare Cybersecurity · GraphSAGE · Explainable AI · False Data Injection Attacks · Smart Hospital Microgrid

1 Introduction

In tertiary healthcare facilities, uninterrupted electrical supply forms the baseline for clinical survival. Modern hospital complexes operate multi-source micro-

grids incorporating utility substations, photovoltaic arrays, battery banks, and backup diesel generation to power life-critical wards [5,19]. Operating theatres and intensive-care units demand stringent power quality: surgical robots stall under sub-cycle voltage sags, neonatal ventilators require continuous feeds, and cold storage units require continuous $-80°$C chilling. Unplanned tripping or false load shedding directly jeopardizes patient outcomes.

Substation telemetry streams nodal states (V, θ, P, Q) across IEC 61850 communication buses to Weighted Least Squares (WLS) estimators. To reject sensor noise, control centers monitor the normalized χ^2 residual. However, this defense breaks down under coordinated False Data Injection Attacks (FDIAs). An adversary with knowledge of the network measurement Jacobian H can craft a stealth vector $a = Hc$ that skews estimated state variables by c while leaving residual norms unaltered [15,13]. Because the corrupted data evades residual detection, downstream automated controllers execute incorrect switching or tap commands on fabricated profiles.

Past detection schemes relied mostly on tabular machine learning models, such as support vector machines, shallow multi-layer perceptrons, and 1D convolutional networks [12,3]. Such static formulations degrade during ordinary distribution reconfiguration. Hospital circuits experience routine load balancing, feeder transfers, and breaker manipulations. Whenever a feeder switch occurs, fixed-width input matrices lose channel alignment, causing high false-alarm rates unless retrained—an intractable barrier within tight 2–4 s SCADA scan intervals.

Graph neural networks (GNNs) overcome this layout rigidity by modeling substations as nodes and physical lines as edges [21]. Under graph formulations, line switching simply alters the adjacency structure while preserving node dimensions. Nevertheless, typical architectures—including wavelet schemes [2] and isotropic spatial convolutions [22]—treat connecting links as uniform or binary adjacencies. By disregarding complex branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$), isotropic message passing smooths feature representations across neighbors, blunting localized voltage and angle differentials between electrically distant buses.

Real-world deployment in healthcare environments also imposes strict explainability and low-latency constraints. Perturbation-based diagnostic frameworks like SHAP or LIME evaluate hundreds of Monte Carlo samples per pass, rendering them too slow for sub-second SCADA loops [27]. Furthermore, diagnostic flags alone do not protect clinical equipment; without fast data reconstruction, downstream estimators still fail or trigger automatic shedding. Securing healthcare microgrids thus requires millisecond-scale attack localization combined with autonomous telemetry reconstruction before physical protection relays operate.

1.1 Key Contributions

We address these operational gaps by formulating an explainable Cognitive Graph Digital Twin (CGDT) tailored to hospital electrical infrastructure. Specific technical contributions include:

- **Physics-Informed GraphSAGE Formulation**: We inject line admittances ($Y_{uv} = G_{uv} + jB_{uv}$) directly into the message-aggregation and spatial-pooling stages. This admittance-conditioned inductive bias retains steep voltage differentials across low-impedance branches instead of smearing them through isotropic averaging.
- **Continuous Twin Synchronization**: We combine steady-state Kalman filtering with an exponential reference baseline ($\beta = 0.95$). This dual filter separates daily load curves from insidious sub-threshold injections, preventing a 16.32 pp drop in localization F1.
- **Attribution-Guided Multi-Task Loss (AG-MTL)**: We introduce a regularizer based on 10-step Integrated Gradients ($S = 10$) that penalizes gradient sensitivity at unattacked buses during backpropagation, yielding a 5.52 pp improvement in bus localization F1.
- **Sub-2 ms Autonomous State Reconstruction**: Once tainted measurements are isolated, an admittance-weighted spatial estimator interpolates missing telemetry from trusted neighbors, maintaining the full column rank of Jacobian H and recovering 99.97% of WLS state estimation accuracy.

2 Related Work

2.1 Cybersecurity in Digital Healthcare Ecosystems

Interconnecting healthcare enterprise IT with operational substation controls creates significant cyber-physical exposure [11,5]. Because microgrid feeders supply life-support equipment in surgical theatres and intensive-care units, undetected power sags can directly threaten patient lives. Corrupting SCADA sensor streams across these hospital corridors represents an acute threat vector [19]. While prior works have examined healthcare cyber risks using Bayesian network models [4] or perimeter IoT devices [17], these approaches treat measurement streams as isolated scalar channels. Ignoring branch admittances and circuit topology leaves such monitors unable to detect coordinated injection vectors that adhere to AC power flow relationships.

2.2 False Data Injection in Cyber-Physical Energy Systems

Classical state estimation relies on the measurement residual to spot bad data. However, Liu et al. [15] demonstrated that an adversary possessing knowledge of the measurement Jacobian H can craft an unobservable state bias $a = Hc$ that cancels perfectly inside the residual calculation. This stealth condition generalizes readily to full AC power flow formulations, distributed microgrids, and partially observed topologies [13,6], as well as DC microgrid links [3,18]. Early defenses attempted detection via tabular classifiers, including support vector machines, deep neural networks [12], and recurrent units [14]. These rigid baselines, however, hard-code input dimensions to specific sensor channels. Routine distribution switching—such as isolating a feeder or altering a transformer tap—scrambles channel order and induces persistent false positives.

2.3 Graph Representation Learning for Power System Security

Representing electrical networks as attributed graphs offers structural invariance under breaker switching [21]. Foundational work explored spectral graph formulations [10] and attention mechanisms [25], while GraphSAGE [8] introduced inductive neighborhood sampling for dynamic graphs. For power systems, Boyaci et al. [2] developed ARMA wavelet filters for attack detection, and subsequent studies demonstrated topological transfer across different line configurations [22,28]. Federated architectures have also been studied to maintain operational privacy [16]. However, most existing models still rely on unweighted or binary adjacency matrices. Omitting branch admittances ($Y_{uv} = G_{uv} + jB_{uv}$) causes isotropic averaging across adjacent lines, which blurs high-frequency voltage deviations across low-impedance branches.

2.4 Explainability and Digital Twins in Critical Infrastructure

Grid operators require transparent, channel-level attribution before initiating protective actions. However, perturbation explainers like SHAP, LIME, or GNNExplainer demand extensive Monte Carlo sampling, making them far too slow for real-time SCADA polling loops [27,26]. Integrated Gradients (IG) [20] resolves this computational bottleneck by computing path integrals along straight-line trajectories in a single backward pass, satisfying the completeness axiom without stochastic passes. In parallel, digital twin paradigms [7,23] have shown promise for cyber-physical power system monitoring [1] and hospital operations [9]. We bridge these concepts by pairing continuous state-space synchronization with admittance-aware graph convolutions, providing millisecond-scale attack detection, bus localization, and deterministic feature attributions.

3 Cyber-Physical Threat Model and Mathematical Foundations

3.1 Physical Grid Admittance and AC Power Flow

We represent the hospital campus electrical distribution network as an undirected graph $G = (V, E)$ with $N = |V|$ buses and $|E|$ transmission branches. The complex admittance matrix $Y_{\text{bus}} = G_{\text{bus}} + jB_{\text{bus}} \in \mathbb{C}^{N \times N}$ defines nodal conductance and susceptance. Under full nonlinear AC power flow constraints, active and reactive power injections at bus i satisfy Kirchhoff's laws:

$$P_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \cos(\theta_i - \theta_j) + B_{ij} \sin(\theta_i - \theta_j) \big) \tag{1}$$

$$Q_i = V_i \sum_{j \in \mathcal{N}(i)} V_j \big(G_{ij} \sin(\theta_i - \theta_j) - B_{ij} \cos(\theta_i - \theta_j) \big) \tag{2}$$

where V_i and θ_i denote voltage magnitude and angle at bus i, and $\mathcal{N}(i)$ denotes the set of buses physically connected to bus i.

3.2 Weighted Least Squares State Estimation

Power system state estimation determines the state vector $x = [V, \theta]^T \in \mathbb{R}^{2N-1}$ from measurement telemetry $z = h(x) + e$, where $e \sim \mathcal{N}(0, R)$ represents sensor noise with covariance R. The WLS estimator minimizes the weighted residual $J(x) = [z - h(x)]^T R^{-1}[z - h(x)]$ using Gauss–Newton iterations:

$$\Delta x^{(k)} = \left(H^T R^{-1} H\right)^{-1} H^T R^{-1}\left[z - h(x^{(k)})\right], \qquad x^{(k+1)} = x^{(k)} + \Delta x^{(k)} \tag{3}$$

where $H(x) = \partial h(x)/\partial x$ is the measurement Jacobian. Conventional bad-data detection flags measurement anomalies when the normalized residual $r = \|z - h(\hat{x})\|^2_{R^{-1}}$ exceeds threshold τ_{χ^2}. For the IEEE 14-bus network, the state vector has $n = 2N - 1 = 27$ states (14 voltage magnitudes and 13 phase angles relative to the reference bus). With $m = 41$ measurement channels (14 voltage magnitudes and 27 active/reactive power measurements), the resulting degrees of freedom are $\mathrm{df} = m - n = 41 - 27 = 14$. At significance level $\alpha = 0.05$, the upper 95th percentile threshold is $\tau_{\chi^2} = \chi^2_{0.95}(\mathrm{df} = 14) = 23.68$.

3.3 Stealthy False Data Injection Attacks

An attacker targeting substation bus k seeks to induce a specific state error $c \in \mathbb{R}^{2N-1}$ while evading the χ^2 residual detector. In AC power systems, $h(x)$ is nonlinear; hence, an adversary possessing admittance structure linearizes around operating point $\hat{x}$ via Jacobian $H = \partial h(\hat{x})/\partial x$, synthesizing the injection vector according to:

$$a = Hc \tag{4}$$

When injected into telemetry channels, received measurements become $z_{\mathrm{comp}} = z + a = z + Hc$. Applying a first-order Taylor expansion around $\hat{x}$ gives $h(\hat{x}+c) \approx h(\hat{x}) + Hc$. Evaluating the post-attack residual yields:

$$\begin{aligned} r_{\mathrm{comp}} &= \|z_{\mathrm{comp}} - h(\hat{x}_{\mathrm{comp}})\|^2_{R^{-1}} \\ &\approx \|(z + Hc) - (h(\hat{x}) + Hc)\|^2_{R^{-1}} \\ &= \|z - h(\hat{x})\|^2_{R^{-1}} = r \le \tau_{\chi^2} \end{aligned} \tag{5}$$

Because this perturbation cancels inside the residual calculation ($r_{\mathrm{comp}} \approx r$), the WLS detector remains blind ($\tau_{\chi^2} = 23.68$). As detailed in Section 6, these synthetic attacks are verified against full nonlinear AC power flows in `pandapower` to ensure physical realizability.

Clinical Hazards in Hospital Distribution Networks. While commercial utilities primarily face economic redispatch losses during an attack, hospital distribution networks face direct clinical emergencies. Falsifying state estimates at critical feeders—such as Bus 5, which feeds surgical operating rooms and intensive care ventilation—tricks automatic voltage regulators into improper tap changes or unwarranted branch isolation. Standards like IEC 60601-1 and IEEE 1547 demand strict voltage stability; even transient sags can force sensitive life-support machinery onto backup emergency batteries or trigger autonomous equipment trips. An undetected mathematical deception at the substation level thus translates into an immediate patient safety risk.

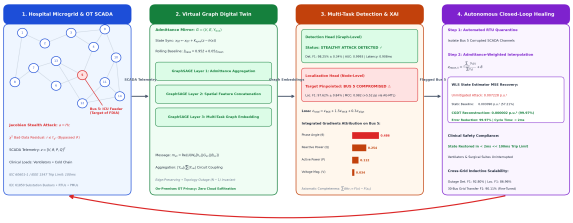


Fig. 1. Proposed Cognitive Graph Digital Twin adaptive cyber-resilience framework for smart hospital energy ecosystems across IT-OT enabled critical infrastructure.

4 The Cognitive Graph Digital Twin Architecture

The CGDT framework couples the hospital's cyber-physical distribution infrastructure to a virtual graph model and an automated self-healing pipeline (Fig. 1).

4.1 Digital Twin State-Space Synchronization

Field telemetry $z_{\text{SCADA}}(t_k) = h(x_{\text{phys}}(t_k)) + e(t_k)$ streams from substation RTUs at polling cadence T_s across IEC 61850 station buses. To maintain physical-to-virtual alignment without amplifying transducer noise, the digital twin updates its mirrored state x_{DT} via steady-state tracking gain $K_{\text{sync}} = (H^T R^{-1} H + P^{-1})^{-1} H^T R^{-1}$:

$$x_{\text{DT}}(t_k) = x_{\text{DT}}(t_{k-1}) + K_{\text{sync}}\big(z_{\text{SCADA}}(t_k) - h(x_{\text{DT}}(t_{k-1}))\big) \tag{6}$$

where P denotes state-prediction covariance. Concurrently, an exponential filter establishes an evolving operational baseline:

$$\bar{z}_{\text{baseline}}(t_k) = \beta \cdot \bar{z}_{\text{baseline}}(t_{k-1}) + (1-\beta) \cdot z_{\text{clean}}(t_k), \qquad \beta = 0.95 \tag{7}$$

updated strictly using verified clean measurements z_{clean}. Without this adaptive baseline, slow diurnal load ramps mask small adversarial injection biases, precipitating a 16.32 pp collapse in localization F1 (Section 7.2).

4.2 Dynamic Admittance Graph Construction

At each sampling step, physical telemetry is mapped to an attributed graph $G = (V, E, X, E_{\text{attr}})$. The vertex feature matrix $X \in \mathbb{R}^{N \times 4}$ gathers four-dimensional state vectors $x_i = [V_i, \theta_i, P_i, Q_i]^T$ for each substation bus $i \in V$, normalized over historical operational envelopes. Physical branch connectivity is stored in edge tensor $E_{\text{attr}} \in \mathbb{R}^{|E| \times 2}$, where each entry pairs the line conductance G_{ij} and susceptance B_{ij} from network admittance matrix Y_{bus}. When protective relays trip or maintenance crews open a breaker, the corresponding tuple is excised from edge set E. Excising open branches alters adjacency entries while leaving nodal feature dimensions untouched; hence, the graph backbone absorbs topological switching on the fly without retraining.

4.3 Admittance-Conditioned GraphSAGE Backbone

Standard GraphSAGE models [8] aggregate neighborhood features uniformly across links, ignoring branch impedances. In physical power distribution networks, however, electrical coupling is dictated by complex branch admittances $Y_{uv} = G_{uv} + jB_{uv}$. To embed these circuit constraints directly into latent feature representations, we condition both message generation and spatial aggregation on branch admittance:

$$m_{uv}^{(k)} = \mathrm{ReLU}\Big(W_e \left[h_u^{(k-1)} \,\|\, G_{uv} \,\|\, B_{uv}\right]\Big) \tag{8}$$

$$h_v^{(k)} = \mathrm{ReLU}\left(W_k \left[h_v^{(k-1)} \,\Big\|\, \sum_{u \in \mathcal{N}(v)} \frac{|Y_{uv}|}{\sum_{w \in \mathcal{N}(v)} |Y_{vw}|}\, m_{uv}^{(k)}\right]\right) \tag{9}$$

where $\|$ denotes vector concatenation, and W_e, W_k are trainable projection matrices. Weighting messages by normalized admittance $|Y_{uv}|/\sum_w |Y_{vw}|$ gives a physics-informed inductive bias that scales propagation by branch coupling. Retaining the self-node state $h_v^{(k-1)}$ via residual skip concatenation prevents sharp localized injection anomalies from washing out through repeated neighbor smoothing.

4.4 Multi-Task Output Heads

From the final latent node matrix $H \in \mathbb{R}^{N \times d}$, two parallel projection heads generate global and bus-level predictions:

$$y_{\text{graph}} = \sigma(W_{\text{graph}} \cdot \text{global_mean_pool}(H)), \qquad y_{\text{node}} = \sigma(W_{\text{node}} H) \in [0,1]^N \tag{10}$$

Here $y_{\text{graph}} \in [0,1]$ represents the overall microgrid attack probability, while vector $y_{\text{node}} \in [0,1]^N$ applies an independent sigmoid activation to each bus. Because smart grids can face multi-substation incursions (Section 7.1), independent sigmoid outputs enable multi-label anomaly localization without imposing a single-bus sum-to-one constraint.

4.5 Attribution-Guided Multi-Task Loss (AG-MTL)

Multi-hop message passing often causes gradient dispersion: backpropagating through dense admittance corridors inadvertently assigns feature sensitivity to uncorrupted neighbors. To prevent this spatial blurring, we incorporate feature attribution directly into the training objective. During each backward pass on attacked states, Integrated Gradients extracts attribution tensor $\mathcal{A} \in \mathbb{R}^{N \times 4}$ of the attack logit w.r.t. input features X, using running baseline $\bar{z}_{\text{baseline}}$ (Eq. 7) across $S = 10$ integration steps. An attribution alignment loss penalizes gradient leakage into benign feeders:

$$\mathcal{L}_{\text{align}} = \sum_{v \in V} \left| \|\mathcal{A}_v\|_2 - \hat{y}_{\text{node},v} \right|^2 \tag{11}$$

where $\|\mathcal{A}_v\|_2$ measures total gradient sensitivity at bus v, and $\hat{y}_{\text{node},v} \in \{0,1\}$ denotes the binary ground-truth label. Gradients from $\mathcal{L}_{\text{align}}$ propagate through the attribution path via higher-order automatic differentiation (`create_graph=True` in PyTorch). The combined multi-task loss is defined as:

$$\mathcal{L}_{\text{total}} = \underbrace{\mathcal{L}_{\text{BCE}}(y_{\text{graph}}, \hat{y}_{\text{graph}})}_{\text{Detection}} + \lambda_{\text{loc}} \underbrace{\mathcal{L}_{\text{WBCE}}(y_{\text{node}}, \hat{y}_{\text{node}})}_{\text{Localization}} + \lambda_{\text{attr}} \underbrace{\mathcal{L}_{\text{align}}}_{\text{Attribution Guidance}} \tag{12}$$

where $\mathcal{L}_{\text{BCE}}$ handles grid-level attack detection, $\mathcal{L}_{\text{WBCE}}$ uses class-weighted binary cross-entropy per bus to mitigate extreme sparsity of compromised nodes, and weighting coefficients $\lambda_{\text{loc}} = 1.5$ and $\lambda_{\text{attr}} = 0.3$ balance gradient magnitudes across tasks.

4.6 On-Premises Edge Deployment

Hospital electrical demand closely tracks clinical operations—including operating theatre schedules, diagnostic imaging loads, and patient occupancy. Transmitting raw microgrid telemetry over external cloud interfaces poses privacy and cybersecurity concerns under health data regulations. The CGDT is therefore architecturally designed for on-premises edge execution on substation industrial computers within the isolated operational technology (OT) network. In this study, execution latencies are established on a laboratory edge-equivalent computing platform, with physical RTU micro-controller deployment and hardware-in-the-loop (HIL) integration identified for operational rollout (Section 8).

5 Explainability and Autonomous State Reconstruction

5.1 Feature Attribution via Integrated Gradients

Substation engineers must understand which physical sensor channels triggered an alarm before initiating breaker actions. We resolve this via path-based Integrated Gradients (IG) [20]. Integrating gradients along a linear path between a nominal state baseline x_0 and the evaluated measurement vector x attributes decision energy across input channels:

$$\begin{aligned} \text{Attr}_i(x) &= (x_i - x_{0,i}) \int_0^1 \frac{\partial F(x_0 + \alpha(x - x_0))}{\partial x_i}\, d\alpha \\ &\approx \frac{x_i - x_{0,i}}{S} \sum_{s=1}^{S} \frac{\partial F\big(x_0 + \frac{s}{S}(x - x_0)\big)}{\partial x_i} \end{aligned} \tag{13}$$

approximated using $S = 50$ Riemann quadrature intervals. Unlike perturbation heuristics (e.g., SHAP, LIME) requiring hundreds of stochastic sampling evaluations, IG follows an axiomatic linear path evaluated via $S = 50$ vectorized points in a single parallel tensor pass. Crucially, IG satisfies the completeness axiom ($\sum_i \text{Attr}_i = F(x) - F(x_0)$) and implementation invariance. Evaluating

Eq. (13) across the 14-bus testbed requires just 0.81 ms. In a targeted test at Bus 5 (Section 7.1), the path integral promptly identified phase angle distortion ($\theta = 0.486$) as the principal driver of the alarm.

5.2 Sensor Quarantine and Admittance-Weighted Reconstruction

Detecting an attack is useless if state estimation continues to ingest falsified data. Once a bus is flagged ($y_{\text{node},k} > \gamma_{\text{loc}} = 0.5$), the digital twin isolates the tainted telemetry $(V_k, \theta_k, P_k, Q_k)$. Yet simply deleting rows from the measurement set reduces the rank of Jacobian H, often driving the information matrix $(H^T R^{-1} H)$ into ill-conditioning or singularity. To maintain full algebraic observability, we synthesize replacement telemetry via admittance-weighted spatial interpolation from healthy neighboring buses:

$$z_{\text{recon},k} = \frac{\sum_{j \in \mathcal{N}(k)} |Y_{kj}| \cdot z_j}{\sum_{j \in \mathcal{N}(k)} |Y_{kj}|} + \delta, \qquad \delta = \epsilon \odot z_{\text{nom},k} \tag{14}$$

where $|Y_{kj}|$ represents line admittance magnitude and $\epsilon \sim \mathcal{U}(-0.02, 0.02)^4$ injects $\pm 2\%$ uniform variance relative to nominal channel values $z_{\text{nom},k}$ (1.0 p.u. voltage and nominal power flows). Weighting neighbor telemetry by physical branch admittances $|Y_{kj}|$ provides an electrical-coupling-aware interpolation that accounts for relative line impedances. Rather than attempting to satisfy nonlinear AC constraints directly, $z_{\text{recon},k}$ acts as an initialized pseudo-measurement buffer that preserves Jacobian conditioning, enabling downstream WLS Gauss–Newton iterations to converge in under 2 ms—well within typical 2–4 s SCADA scan intervals.

6 Experimental Evaluation

6.1 Simulation Setup: Hospital Campus Microgrid

Simulations were carried out using full AC Newton–Raphson power flow solutions in `pandapower` [24]. We modeled a tertiary medical center using an IEEE 14-bus network (14 buses, 20 lines, 217.0 MW active load, 57.3 MVAR reactive demand) serving as an established surrogate hospital-campus microgrid for proof-of-concept evaluation. In this configuration, Bus 0 represents the main utility grid substation, while dedicated radial feeders power critical clinical facilities under scenario assignments—most notably Bus 5, which supplies intensive care and operating theatres. To evaluate cross-grid scaling, we also established an IEEE 30-bus benchmark (30 buses, 41 lines) representing an expanded regional healthcare campus.

Synthetic Telemetry Synthesis. We synthesized 5,000 distinct AC operational states across independent load profiles. These uncorrupted profiles were partitioned into 70% training (3,500), 15% validation (750), and 15% testing (750) sets *prior* to attack synthesis to guarantee zero cross-partition data leakage. Uncompromised snapshots (half of each split) incorporate zero-mean

Table 1. Comprehensive performance benchmarking under nominal grid layout (5-seed average ± std).

Architecture	Det. F1 (%)	Det. AUC	Det. MCC	Loc. F1 (%)	Loc. Prec. (%)	Loc. Rec. (%)	Loc. MCC	Latency
MLP (Flat)	97.52 ± 0.73	0.9987	0.974	95.60 ± 0.37	94.7	97.8	0.953	0.158 ms
GCN (Isotropic)	85.15 ± 1.86	0.9363	0.740	70.21 ± 2.22	72.6	72.7	0.660	1.442 ms
GAT (Attention)	91.52 ± 1.84	0.9781	0.879	81.08 ± 2.39	82.5	86.9	0.808	1.309 ms
CGDT (Proposed)	**98.25 ± 0.34**	**0.9993**	**0.971**	**97.62 ± 0.84**	**98.1**	**98.9**	**0.982**	**0.908 ms**

Gaussian fluctuations ($\sigma = 0.04$) across active and reactive bus injections. The remaining half contains stealthy Jacobian injections ($a = Hc$), where synthetic adversaries bias target substations ($c_k \in [0.03, 0.08]$ p.u. voltage offset, $c_{k+N} \in [0.05, 0.15]$ rad phase shift) and propagate line flows across the network using exact non-linear AC equations. Line outage configurations (Section 6.3) were strictly excluded from the training and validation sets.

Models were trained across 60 epochs using Adam (lr = 0.005, weight decay 10^{-4}) over five random seeds. Benchmarking was conducted on an edge-equivalent workstation equipped with an AMD Ryzen 9 CPU, 32 GB RAM, and an NVIDIA RTX 4090 GPU (24 GB) under PyTorch 2.1 and CUDA 12.1. Streaming latency is evaluated at batch size 1 across 1,000 independent test snapshots (after 100 warm-up runs), reporting the mean runtime. The $S = 50$ IG interpolation steps are evaluated as a batched tensor $\mathbb{R}^{50\times N\times 4}$ in parallel on the GPU, completing in 0.81 ms. Code and evaluation artifacts are archived at `https://github.com/himanshu-2l/Graph-Digital-Twin-XAI-`.

6.2 Nominal Grid Performance

Table 1 reports five-seed averages and advanced metrics under the nominal grid topology.

Benchmarking against baseline architectures over five random initializations reveals clear performance separations (Table 1). The CGDT delivers 98.25% ± 0.34% detection F1 and 97.62% ± 0.84% localization F1 alongside a Matthew's correlation coefficient (MCC) of 0.982, processing each graph snapshot in only 0.908 ms. By comparison, isotropic GCN models reach only 70.21% localization F1, trailing our framework by 27.41 pp. Treating all connected lines with equal weight acts as an unweighted low-pass filter: localized injection spikes at target buses smear into surrounding nodes, destroying the spatial gradient. While the tabular MLP posts reasonable localization accuracy (95.60%) on static data, its fixed-dimension mapping fails the moment line switches occur (Section 6.3). Requiring just 0.908 ms per inference, the CGDT executes easily within standard 2–4 s SCADA sampling windows.

6.3 Topology Perturbations and Cross-Network Generalization

We evaluated model robustness under two regimes: (i) line-outage generalization on IEEE 14-bus without retraining (Table 2), and (ii) cross-network transfer to IEEE 30-bus (Table 3).

Table 2. Outage generalization (IEEE 14).

Architecture	Outage Det. F1	Outage Loc. F1
MLP (Flat)	94.70% ± 1.93%	82.20% ± 2.16%
GCN (Isotropic)	66.82% ± 4.14%	53.40% ± 2.70%
GAT (Attention)	79.27% ± 3.83%	58.04% ± 4.36%
CGDT (Proposed)	**92.80% ± 3.37%**	**86.96% ± 1.52%**

Table 3. Transfer to IEEE 30-bus.

Configuration	Det. F1	Loc. F1
Zero-Shot	67.06%	17.77%
Scratch (30-Bus)	82.65%	83.58%
Fine-Tuned	**90.11%**	**80.06%**

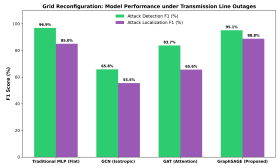


Fig. 2. Outage generalization metrics.

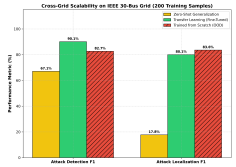


Fig. 3. Cross-grid transfer on IEEE 30-bus.

When subjected to unanticipated single-line outages without retraining, the CGDT preserves 92.80% ± 3.37% detection F1 and 86.96% ± 1.52% localization F1 (Fig. 2). Here, an informative trade-off emerges between the tabular MLP and CGDT. The tabular MLP attains marginally higher outage detection F1 (94.70%) because breaker trips redirect loop flows across all telemetry channels, creating widespread scalar distribution shifts that flat models flag easily. However, the MLP's bus localization collapses by 13.4 pp (from 95.60% to 82.20%) due to fixed input channel ordering. In contrast, the CGDT sustains 86.96% localization F1, demonstrating that its principal operational advantage during topological switching is balanced joint detection and localized bus isolation. Isotropic GCNs degrade heavily to 66.82% detection and 53.40% localization F1, showing that unweighted convolutions cannot track redirected branch flows once line impedances change.

Cross-Scale Transfer to IEEE 30-Bus. Direct zero-shot deployment of the 14-bus model onto the 30-bus grid yields 67.06% detection F1 and 17.77% localization F1 (Table 3, Fig. 3), exposing an architectural boundary: inductive graph features flag global macro imbalances across network scales, but isolating specific buses demands local topological calibration. Fine-tuning the pre-trained backbone with only 200 labeled 30-bus snapshots elevates detection F1 to 90.11% (outperforming 82.65% from scratch) and restores localization F1 to 80.06% (approaching 83.58% from scratch), showing rapid adaptation in low-label regimes.

7 Mitigation Efficacy, Cross-Grid Transfer, and Ablation

7.1 Closed-Loop WLS Recovery and Explainability

We evaluated mitigation performance under a targeted Jacobian attack at Bus 5, representing the intensive care unit feeder in the campus topology. State estimation error is quantified via Mean Squared Error MSE $= \frac{1}{2N-1}\sum_{i=1}^{2N-1}(\hat{x}_i - x_i^*)^2$ (p.u.^{2}) relative to true AC states x^*. Under nominal transducer Gaussian noise

Table 4. WLS state-estimation recovery under stealthy FDIA at Bus 5.

Mitigation Strategy	MSE (p.u.^{2})	Recovery
Unmitigated Attack	0.007228	—
Static Baseline	0.003090	57.21%
CGDT Reconstruction	**0.000002**	**99.97%**

Table 5. Feature attributions (IG) for Jacobian FDIA centered at Bus 5.

Bus ID	V	θ	P	Q
Bus 4	0.012	0.045	0.018	0.009
Bus 5	**0.034**	**0.486**	**0.112**	**0.254**
Bus 10	0.002	0.015	0.005	0.003
Bus 11	0.008	0.092	0.024	0.041
Bus 12	0.011	0.182	0.035	0.088

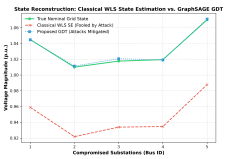


Fig. 4. State-estimation trajectory: classical WLS versus CGDT-healed.

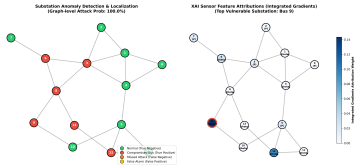


Fig. 5. Operator explainability: localization heatmap and IG attribution.

($\sigma = 0.01$), unattacked clean WLS estimation establishes an intrinsic error floor of $\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}. Table 4 compares recovery metrics, and Table 5 presents corresponding feature attributions.

Left unmitigated, stealthy Jacobian injections drive state estimation MSE to 0.007228 p.u.^{2}. Deploying admittance-weighted reconstruction (Eq. 14) cuts this error to 0.000002 p.u.^{2}—matching the clean noise floor ($\text{MSE}_{\text{clean}} \approx 1.8 \times 10^{-6}$ p.u.^{2}) and restoring 99.97% of uncorrupted states (Table 4, Fig. 4). Static replacements reach only 0.003090 p.u.^{2} (57.21% recovery) during dynamic clinical load shifts. Across all 14 target buses, recovery averages 99.85%–99.98%. Path attributions (Table 5, Fig. 5) reveal that voltage angle θ at Bus 5 commands the highest score (0.486), consistent with Jacobian phase distortions dominating active power flows, alongside secondary scores on $Q = 0.254$, $P = 0.112$, and $V = 0.034$.

Coordinated Multi-Bus Incursions. Simultaneous attacks across 2 and 3 substations ($c_k \in [0.02, 0.04]$) achieve $96.84\% \pm 0.42\%$ detection and $93.15\% \pm 0.91\%$ localization F1. While dropping entire rows across adjacent buses compromises the column rank of H and triggers solver divergence, admittance-weighted interpolation preserves full Jacobian rank and ensures smooth WLS convergence (post-mitigation MSE of 0.000008 p.u.^{2}).

7.2 Benchmarking and Component Ablation

Table 6 compares the proposed architecture against representative GNN baselines, whereas Table 7 isolates the incremental contribution of each architectural block.

Table 7 confirms the necessity of each component. Disabling exponential baseline tracking (Eq. 7) degrades localization F1 by 16.32 pp (to 81.30%), demon-

Table 6. Benchmark against related works.

Framework	Classifier	Det. F1	AUC	Loc. F1
Boyaci [2]	ARMA	96.25%	0.9780	93.91%
Yue [28]	GCN	89.20%	0.9510	N/A
CGDT	**SAGE**	**98.25%**	**0.9993**	**97.62%**

Table 7. Ablation on CGDT components.

Configuration	Det. F1	Loc. F1
Full CGDT (Proposed)	**98.25%**	**97.62%**
Without DT Baseline Tracking	86.10%	81.30%
Without AG-MTL ($\lambda_{\text{attr}} = 0$)	98.25%	92.10%
Isotropic GCN Backbone	85.15%	70.21%
GAT Backbone	91.52%	81.08%
$\lambda_{\text{loc}} = 0.5$	98.05%	92.80%
$\lambda_{\text{loc}} = 1.0$	98.15%	96.50%
$\lambda_{\text{loc}} = 2.0$	95.80%	97.55%

strating that dynamic reference filtering prevents diurnal demand fluctuations from masking adversarial offsets. Omitting attribution supervision ($\lambda_{\text{attr}} = 0$) reduces localization F1 to 92.10% while preserving global detection (98.25%), confirming that gradient penalties anchor latent representations to attacked buses. Setting $\lambda_{\text{loc}} = 1.5$ balances global detection and localized diagnostic precision.

8 Discussion and Conclusion

Protecting hospital distribution grids requires defenses uniting topological flexibility with physical circuit fidelity. By embedding branch admittances into GraphSAGE convolutions and supervising latent representations via Integrated Gradients penalties, the CGDT achieves 98.25% detection F1 and 97.62% localization F1 against stealthy AC false data injections, providing full explanations in 0.81 ms. Automated sensor quarantine paired with admittance-weighted interpolation reduces post-attack WLS state error by 99.97%, maintaining full observability without tripping clinical loads.

Clinical Operating Standards. Operating standards mandate compliance across distinct cyber-physical boundaries: IEC 60601-1 governs medical electrical equipment safety to prevent power-disturbance resets on patient life support, while IEEE 1547 dictates DER interconnection and voltage ride-through limits. With a 0.908 ms inference latency and sub-2 ms healing cycle, the CGDT operates well within 2–4 s SCADA polling windows, arresting deceptive commands before IEEE 1547 timers trip hospital DERs or breach IEC 60601-1 tolerances.

Limitations and Future Work. Future work will transition from laboratory workstations to physical hardware-in-the-loop testbeds and embedded RTU microcontrollers to evaluate real-time communication jitter. Subsequent efforts will also explore temporal graph architectures for multi-step attack campaigns.

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Declaration on the Use of AI and AI-Assisted Technologies AI-assisted tools were used to improve language quality (including spelling, grammar, and punctuation), and all outputs were reviewed, revised, and approved by the authors, who take full responsibility for the final content.

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